

STEPHENSON'S ROCKET

In 1829, five locomotives competed before a huge crowd at Rainhill, Lancashire, UK, to decide which would haul carriages on the world's first passenger railroad, between Liverpool and Manchester in the UK. The winner was the British engineer Robert Stephenson's *Rocket*, which at one point reached a speed of 25 mph (40 km/h). It soon became the world's most famous locomotive.

Chimney releases hot air and smoke via the boiler.

WOW!

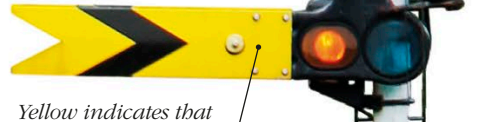
Within 45 years of the *Rocket*, over 160,000 miles (257,000 km) of railroad track had been built around the world.

Red, square-ended horizontal arm means "stop."



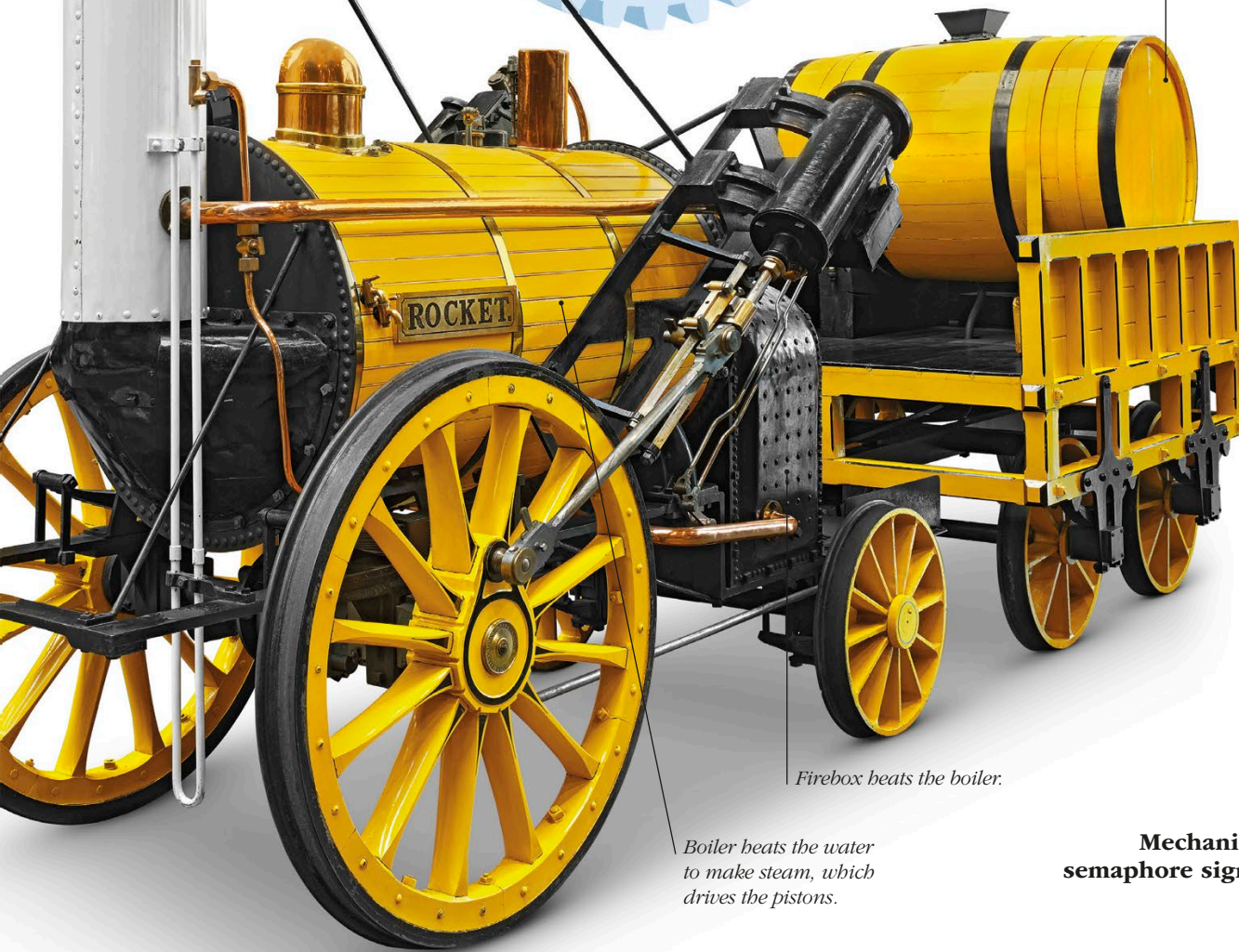
WARNING SIGNALS

Semaphore signals are hinged arms that can be set at different angles to send messages to a train driver. First used in 1842 by the British engineer Charles Hutton Gregory, semaphore signals warned train drivers about hazards ahead. They have been replaced by colored light signals over the years.



Yellow indicates that this is a distant signal and the horizontal arm indicates "caution."

Water barrel supplies water to the boiler.



Firebox heats the boiler.

Boiler heats the water to make steam, which drives the pistons.

Mechanical semaphore signal